Philosophy Of Engineering

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/Important Topics & Exam Pattern.url =====
CATEGORY: PYQs
STATUS: url_reference_only

URL: https://docs.google.com/document/d/1WGW5jJVAoTk7598I9wLGGUVpJNds1__W7QIiqOlPEl8/edit?usp=sharing

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/MCQ & Question Bank.url =====
CATEGORY: PYQs
STATUS: url_reference_only

URL: https://drive.google.com/drive/folders/1um6ctu9sT624qLw5U1SD0PL1OETqmSCM?usp=drive_link

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/MCQ Test 2025.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
Philosophy of Engineering (POE) – CT1 (All 20
Questions with Options and Answers)
Title: Philosophy of Engineering (POE) - CT1 (1 to 20)
1. Define Engineering:
a) Study of natural laws
b) Application of science, mathematics, and technology to solve problems
c) Understanding of society
d) Philosophical analysis
Answer: b) Application of science, mathematics, and technology to solve problems
2. A team compares how water systems were managed in ancient villages, medieval castles, and modern smart cities.
a) Scientific Inquiry
b) Historical comparison of engineering development
c) Technological Impact Assessment
d) Philosophical Reflection
Answer: b) Historical comparison of engineering development
3. Name the five elements of STEAM.

a) Science, Technology, Engineering, Arts, Mathematics
b) Science, Technology, Economics, Arts, Management
c) Statistics, Technology, Engineering, Arts, Management
d) Science, Technology, English, Arts, Mathematics
Answer: a) Science, Technology, Engineering, Arts, Mathematics
4. In a class, students debate whether adding music, design, and aesthetics improves engineering
problem-solving.
a) STEAM Pyramid – Art context
b) Attributes of Engineers
c) Industrial Age Innovations
d) Social Development Theories
Answer: a) STEAM Pyramid – Art context
5. Which is NOT a desired attribute of an engineer?
a) Logical Thinking
b) Communication
c) Lack of adaptability
d) Problem Solving
Answer: c) Lack of adaptability
6. An engineer is recruited for creativity, persistence, and teamwork skills. Which framework
describes these qualities?
a) Engineering Habits of Mind
b) STEAM Pyramid
c) Medieval Engineering History
d) Industrial Economy framework
Answer: a) Engineering Habits of Mind
7. Find the subject linking science and technology with engineering.
a) Mathematics
b) Philosophy

c) Arts
d) History
Answer: a) Mathematics
8. A group presents on Roman bridges, medieval aqueducts, and AI-enabled smart bridges.
a) Historical comparison of engineering development
b) Engineering Design Methodology
c) Ontological Study
d) Analytical Design
Answer: a) Historical comparison of engineering development
9. Which of the following is NOT in STEAM?
a) Science
b) Mathematics

c) Economics
d) Engineering
Answer: c) Economics
10. A report highlights an engineer who is innovative, ethical, and a successful leader. Which
practice evaluates such attributes?
a) Historical Comparison
b) STEAM Pyramid
c) Case Study on Attributes of Engineers
d) Habits of Mind framework
Answer: c) Case Study on Attributes of Engineers
11. Why has engineering evolved historically?
a) For political influence
b) For solving problems with science and technology
c) For entertainment
d) For arts promotion
Answer: b) For solving problems with science and technology
12. Label the role of philosophy of engineering.
a) To improve coding practices
b) To understand engineering’s foundation and its social role
c) To study artistic concepts
d) To calculate efficiency
Answer: b) To understand engineering’s foundation and its social role
13. Name the ancient structure often cited as an example of early engineering philosophy.
a) Great Wall of China
b) Pyramids of Giza
c) Stonehenge
d) Colosseum
Answer: b) Pyramids of Giza
14. What does Ontology in Engineering primarily deal with?
a) Study of the physical properties of materials
b) Study of the existence and relationships of engineering concepts
c) Study of economic feasibility of projects
d) Study of mathematical optimization methods
Answer: b) Study of the existence and relationships of engineering concepts
15. Product Life Cycle Ontology advantage for EVs:
a) It focuses on marketing and sales only

b) It integrates processes across different life cycle stages
c) It improves coding efficiency
d) It avoids recycling processes
Answer: b) It integrates processes across different life cycle stages
16. Which of the following best describes Reference Ontology?

a) A set of informal practices engineers use
b) A universal framework describing common engineering concepts
c) A tool for financial planning
d) A system for coding programming languages
Answer: b) A universal framework describing common engineering concepts
17. Application Ontology focuses on:
a) Domain-specific implementation of engineering concepts
b) Historical analysis of engineering development
c) General attributes of an engineer
d) The STEAM Pyramid
Answer: a) Domain-specific implementation of engineering concepts
18. Smart city ontology approach:
a) Case-based Study
b) Reference Ontology with Concept Mapping
c) Historical Analysis
d) Mathematical Ontology
Answer: b) Reference Ontology with Concept Mapping
19. Which method is often used in building Ontologies in Engineering?
a) Data Tables
b) Concept/Mind Mapping
c) Empirical Research
d) Flowcharting
Answer: b) Concept/Mind Mapping
20. An engineer is tasked with developing a domain-specific ontology for railway systems (signals,
stations, tracks). Which type of ontology is most suitable?
a) Reference Ontology
b) Application Ontology
c) Flowchart Ontology
d) Mathematical Ontology
Answer: b) Application Ontology

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/PYQ Jan 2023.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
Reg. No. B2oO3PT74
B.Tech./ M. Tech (lntegrated) DEGREE EXAMINATION, JANUARY 2023 First Semester
21GNH101J-PHIL.OSOPHY OF ENGINEERING (For the candidates admitted firom the academic year 2022-2023) Note: deo artA Should de answered in OMR sheet within first 40 minutes and OMR sheet should be na
over to hall invigilator at the end of 40th minute. Part-Band Part-C should be answered in answer booklet. (i) (i) Time: 3 Hours Max. Marks: 75

Mark nI, CO ro
PART-A (20 x 1 -20Marks) Answer ALL Questions cra . Mathematics and science were applied in all the ficlds in .
(A) Ancient era (C) Renaissance era (B) Middle era () Modern era
2 engineering developed during the industrial revolution.
(A) Mechanical (C) Aeronautical (B) Chemical (D) Electrical
3. can be viewed as an activity that forms or changes culture.
(A) Science (C) Arts (B) Engineering (D) Technology
oriented process of designing and making tools 4. Enginecering is the and systems. (A) Science (C) Design B) Goal (D) Technology 2 2 is the branch of philosophy that studies concepts such as 5. existence, being and becoming. (A) Ontology (C) Engineering (B) First-order logic (D) Axiology 1 21 identifies ontology's function with respect to its accuracy and 6. comprehensiveness. (A) Quantity (C)Weakness (B) Strength (D) Quality 7. Which of the following ontology is specific-domain independent? (A) Foundational (C) Domain
22 (B) Reference (D) Application 8. Choose the last stage of product life cycle from the options listed below. (A) Product maturity (C) Product decline
2 (B) Product growth (D) Product development Page 1 of 3 07JF1-21GNH101J

[Page 2]
is the total of all engineered tools, devices and processes available (B) Engineering (D) Knowledge le. 1 9 (A) Technology (C) Science 10. In which quadrant, social sciences fall-engineer as (A) Scientist (C) Doer 23 (3) Sociologist D) Designer 11. Persuaders are active in (A) Artistic G)Enterprising phase of RIASEC model. (B) Social D) Investigative 12. A division of epistemology which is crucial to develop scientific initiatives1 is called 31 (A) Design epistemology C) Activity epistemology (B) Planning epistemology (D) Timing epistemology 241 method asks for a question to the user or a person. (A) Scientific C)Technical 13. (B) Engineering D) Research 14. If the objective of your project is to invent a new product or environment, then (A) Scientific (C) Technology method you will follow? (B) Technical D) Engineering 114 1 model is called as instructional systems design. (A) RAISEC model (C) Scientific model 15. (B) ADDIE model D) Engineering model 16. Transformation of design into product falls under stage of CDIO 1 142 process. (A) Conceive C)Implement B) Design (D) Operate field there exists minimum contribution from engineering and 17. In society. (A) Health C) Space 124 1 (B) Water (D) Modern homes should hold paramount of safety, health and welfare of the public. (A) Engineers C) Social welfare 18.
114
(B) Artistics D) Innovators 19. 3Es stand for I4
1 (A) Ethics, equality and economics (B) Economics, environmental and equality C) Equality, environmental and () Environmental, ethics and ethics economics

[Page 3]
20. Which one of the listed engineering associations deal with puo1a conference? (A) National society of engineers (B) IEEE (C) Society of women engineers (D) ISCA
Marks BL CO PO
PART B (4 x 10 40 Marks) Answer ANY FOUR Questions 10 1 21. Explain in detail about the relationship between arts, mathematics, science, technology and engineering, 10 1 1I 22.

Draw the STEAM pyramid and explain its components. 2 1 23.i. Briefly explain the ontological layers with neat sketch. 2 ii. List the difference between ontologies with neat table. 5 13 24.i. State the definition and difference between science, engineering and technology. 5 13 ii. Explain the four dimensions of engineering with neat sketch. 10 4 1 25. State the difference between scientific method and engineering design with neat diagram. 10 5 1 1 26. Brief the aspects of 3E's that could lead to sustainable development.
Marks BL CO PO PART C(1x15=15 Marks) Answer ANY ONE Questions
15 5 4 3 27. Create one course as an illustration. Write down the significance of each stages of the ADDIE model in the course. How do you map various stages of ADDIE model with teaching learning process? Do you believe that the course would have been more beneficial if you had included the stages which are missed? Mention and defend the stages that you believe should be included. Do you believe the outcomes would be worth the effort given the amount of work required to move through each stage? Justify. 28. In order to find new drugs, XYZ pharmaceuticals is doing research and 15 5 experiments. Think about this scenario and list out the challenges related to science, engineering and technology. Describe each challenges and figure out the possible solutions. Decide whether science or engineering or technology is more appropriate in this situation. Justify your choice.
* ***

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/PYQ May 2023.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
31MF1/2-21GNH1013
(C) As decided by manufacturer
(A) Higher
(D) Lower
selling price will be
8. When a product is in its decline stage of its product life cycle, the average
(B) Average
1
1 2 1
(C) Product maturity (A) Product development
(D) Product decline
7. What happens in the first stage of product life cycle? (B) Product growth
1 1 2 1
(C) Theoretical (A) Reference
(D) Metaphysical
domain and task
(B) Application.
6
ontologies describes concepts depending both on a particular

1 2 1
(C) Ontology (A) Physiology 5
belongs to the major branch of metaphysics. (D) Sociology (B) Biology
1 1 2
(C) 1998 (A) 1947 4. The frst steam engine was built in (D) 1647 (B) 1698
1 1 1 1
(C) 1174 and 1200
(A) 1114 and 1200
(D) 1074 and 1200
3. Modern ERA is between
(B) 1104 and 1200
1
(C) English
1 1 1
(A) Greek
(D) Arabic
2. The term engine is derived from (B) Latin
(C) Teotihulan
1 1 1 1
(A) Imhotep
(D) Saqqara
1. The earliest Civil Engineer is known by name (B) Babylon
Answer ALL Questions
1
1 1 1
PART-A (20 x 1 = 20Marks)
Marks BL
CO PO
Time: 3 Hours
Max. Marks: 75
(ü) Part -B and Part -C should be answered in answer booklet.
i) over to hall invigilator at the end of 40th minute.
Part -A should be answered in OMR sheet within frst 40 minutes and OMR sheet should be handed
Note:
(For the candidates admitted from the academic year 2022-2023)
21GNHI01J
-PHILOSOPHY
OF ENGINEERING
First/ Second Semester
B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION,
MAY 2023
Reg. No. RA2
O37
2 1

Paee
(C) Doctors
(D) Engineers (B) Teachers
(A) Actors 20.
are involved in planning and managing
projects
1 2 5I
(C) Equity (A) Sustainability compromising
the ability of future generations
to meet their needs
19. Development
which meets the needs of current generations
without
(D) Integrity (B) Diversity
1 1
(C) 1897 (A) 1997
(D) 1889 (B) 1979.
1 2 5 1
(C) Ethical
(D) Equity
(A) Environment
(B) Economic
1
17.
is the most discussed
aspect of sustainability.
(C) Development phase (A) Evaluation phase
14
16. The prototype creation is involved in (D) Design phase (B) Implementation phase phase of addie model.
(C) Communicate results
(D) Establish constraints
4
(A) Define problem 15. The final stage of engineering
design process is (B) Research ideas
1
(C) 6 (A) 2
(D) 8 (B) 4
1
14.
number of parts are involved in evaluation
phase.
(C) Holland code

(D) Engineering method
13. The (A) Scientific method
test your hypothesis
by doing an experiment (B) Addie model
3 2
(C) Thinkers
(D) Doers (B) Helpers
1
12. (A) Creators focus on ideas
2
(C) 6" (A) 2 11, Holland's theory describes
(D) 8 (B) 4 number of basic personality
types.
3
(C) Execution
(D) Epistemology
products 10. Design as (A) Activity
(B) Planning
1
(C) Technology
is related to the conceptualization
stages of making
new
(A) Engineering
(B) Social (D) Science
9.
is the study of the natural world as it is
51
18. The American
Association
of Engineering
Societies
was established
in

28. Give detailed notes on RAISEC model.
15
1 2
27. What do you mean by product life cycle? Explain its stages in detail.
15
1 2
Answer ANY ONE Question
PART -C(1×
15 =15 Marks)

Marks
BL CO PC
26. Mention the professional
organizations
engineers
detailed note. 25. What do you mean by CDIO engineers
in industry?
Explain in detail.
10
1 5 1
10
1 4
method and engineering
design.
1
between scientific
10
1 2 1
23. What are the four dimensions
dimensions. 22. Explain reference ontology with
10
1 3 1
21. Explain briefly on
regard to application
ontology.
10
1 2 1
non-motivated functions.
10
1 1 1
Answer
ANY FOUR Questions
PART -B (4 × 10 = 40
Marks
BL CO PO
available
with a for
24. Differentiate
of engineering?
Write short notes on these
Marks)

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/pyq txt/MCQ Test 2025.txt =====
CATEGORY: PYQs

STATUS: ok

Philosophy of Engineering (POE) – CT1 (All 20
Questions with Options and Answers)

Title: Philosophy of Engineering (POE) - CT1 (1 to 20)

1. Define Engineering:
a) Study of natural laws
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b) Science, Technology, Economics, Arts, Management
c) Statistics, Technology, Engineering, Arts, Management
d) Science, Technology, English, Arts, Mathematics
Answer: a) Science, Technology, Engineering, Arts, Mathematics

4. In a class, students debate whether adding music, design, and aesthetics improves engineering
problem-solving.
a) STEAM Pyramid – Art context
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d) Industrial Economy framework
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7. Find the subject linking science and technology with engineering.
a) Mathematics
b) Philosophy

c) Arts
d) History
Answer: a) Mathematics

8. A group presents on Roman bridges, medieval aqueducts, and AI-enabled smart bridges.
a) Historical comparison of engineering development
b) Engineering Design Methodology
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b) Mathematics
c) Economics
d) Engineering
Answer: c) Economics

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practice evaluates such attributes?
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c) Case Study on Attributes of Engineers
d) Habits of Mind framework
Answer: c) Case Study on Attributes of Engineers

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a) For political influence
b) For solving problems with science and technology
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d) For arts promotion
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12. Label the role of philosophy of engineering.
a) To improve coding practices
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d) It avoids recycling processes
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b) A universal framework describing common engineering concepts
c) A tool for financial planning
d) A system for coding programming languages
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17. Application Ontology focuses on:
a) Domain-specific implementation of engineering concepts
b) Historical analysis of engineering development
c) General attributes of an engineer
d) The STEAM Pyramid

Answer: a) Domain-specific implementation of engineering concepts

18. Smart city ontology approach:
a) Case-based Study
b) Reference Ontology with Concept Mapping
c) Historical Analysis
d) Mathematical Ontology
Answer: b) Reference Ontology with Concept Mapping

19. Which method is often used in building Ontologies in Engineering?
a) Data Tables
b) Concept/Mind Mapping
c) Empirical Research
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20. An engineer is tasked with developing a domain-specific ontology for railway systems (signals,
stations, tracks). Which type of ontology is most suitable?
a) Reference Ontology
b) Application Ontology
c) Flowchart Ontology
d) Mathematical Ontology
Answer: b) Application Ontology

===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/pyq txt/PYQ Jan 2023.txt =====
CATEGORY: PYQs
STATUS: ok

Reg. No.B2oO3PT74

B.Tech./ M. Tech (lntegrated) DEGREE EXAMINATION, JANUARY 2023

First Semester

ERING
21GNH101J- PHIL.OSOPHY OF ENGINE

2023)
(For the candidates admitted firom the academic year 2022-

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Note: d be na

t 40 minutes and OMR sheet shoul

(i) artA Should de answered in OMR sheet within firs

over to hall invigilator at the end of 40th minute.

(i) Part- Band Part-C should be answered in answer booklet.
Max. Marks: 75

Time: 3 Hours ro
nI, CO

Mark

PART-A (20 x 1 - 20Marks)
Answer ALL Questions

cra
in all the ficlds in

.

. Mathematics and science were applied
(B) Middle era

(A) Ancient era
() Modern era

(C) Renaissance era

trial revolution.

2 engineering developed during the indus

(B) Chemical
(A) Mechanical (D) Electrical
(C) Aeronautical

at forms or changes culture.
viewed as an activity th

3. can be (B) Engineering
(A) Science (D) Technology
(C) Arts

nd making tools

4. Enginecering is the oriented process of designing a

and systems. B) Goal
(A) Science (D) Technology
(C) Design 2 2

tudies concepts such as

is the branch of philosophy that s

5. .
existence, being and becoming (B) First-order logic
(A) Ontology (D) Axiology
(C) Engineering

1 21
cy and

identifies ontology's function with respect to its accura

6.
comprehensiveness.

(B) Strength
(A) Quantity

(D) Quality
(C)Weakness

22
main independent?

7. Which of the following ontology is specific-do
(B) Reference

(A) Foundational
(C) Domain (D) Application

2
8. Choose the last stage of product life cycle from the options listed below.

(A) Product maturity (B) Product growth
(C) Product decline (D) Product development

9 is the total of all engineered tools, devices and processes availabllee. 1

(A) Technology (B) Engineering
(C) Science (D) Knowledge

10. In which quadrant, social sciences fall-engineer as

(A) Scientist (3) Sociologist 23
(C) Doer D) Designer

11. Persuaders are active in phase of RIASEC model.

(A) Artistic (B) Social

G)Enterprising D) Investigative

12. A division of epistemology whichis crucial to develop scientific initiatives1
31

is called
(A) Design epistemology (B) Planning epistemology
C) Activity epistemology (D) Timing epistemology

13. method asks for a question to the user or a person. 241
(A) Scientific (B) Engineering
C)Technical D) Research

14. Ift he objective of your project is to invent a new product or environment,
then method you will follow?
(A) Scientific (B) Technical
(C) Technology D) Engineering

15. model is called as instructional systems design. 114 1
(A) RAISEC model (B) ADDIE model
(C) Scientific model D) Engineering model

16. Transformation of design into product falls under stageo f CDIO 1 142

process.
(A) Conceive B) Design
C)Implement (D) Operate

17. In field there exists minimum contribution 124 1
society. from engineering and

(A) Health
(B) Water

C) Space (D) Modern homes
18. should hold paramount of 114

(A) Engineers safety, health and welfare of the
C) Social welfare (B) Artistics public.

D) Innovators
19. 3Es stand for

I4

(A) 1
Ethics, equality and economics (B) Economics, environmental and

C) Equality, environmental and equality
()

ethics Environmental, ethics and
economics

20. Which one of the listed engineering associations deal with puo1a
conference?
(A) National society of engineers (B) IEEE

(C) Society of women engineers (D) ISCA

Marks BL CO PO

PART B (4 x 10 40 Marks)
Answer ANY FOUR Questions

10 1

21. Explain in detail about the relationship between arts, mathematics, science,

technology and engineering,
10 1 1I

22. Draw the STEAM pyramid and explain its components.
2 1

23.i. Briefly explain the ontological layers with neat sketch.
2

ii. List the difference between ontologies with neat table.

5 13

24.i. State the definition and difference between science, engineering and

technology.
5 13

ii. Explain the four dimensions of engineering with neat sketch.
10 4 1

25. State the difference between scientific method and engineering design with

neat diagram.
10 1 5 1

26. Brief the aspects of 3E's that could lead to sustainable development.

Marks BL CO PO
PART C(1x15=15 Marks)

Answer ANY ONE Questions

27. Create one course as an illustration. Write down the significance of each 15 5 4 3

stages of the ADDIE model in the course. How do you map various stages
of ADDIE model with teaching learning process? Do you believe that the
course would have been more beneficial if you had included the stages
which are missed? Mention and defend the stages that you believe should
be included. Do you believe the outcomes would be worth the effort given
the amount of work required to move through each stage? Justify.

28. In order to find new drugs, XYZ pharmaceuticals is doing research and 15 5

experiments. Think about this scenario and list out the challenges related to
science, engineering and technology. Describe each challenges and figure
out the possible solutions. Decide whether science or engineering or

technology is more appropriate in this situation. Justify your choice.

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===== SOURCE_FILE: nexus sem/Philosophy Of Engineering/PYQs/pyq txt/PYQ May 2023.txt =====
CATEGORY: PYQs
STATUS: ok

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